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"Most Important"

"Algebra"

Sequences & Series

Grade 11 • Mathematics • Module 1

"Progressions"







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1 n^{th} term of a Sequence

2 Arithmetic Progression

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4 Arithmetic Geometric Progression

5 Harmonic Progression and Relation between A.M and G.M

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1. n^{th} term of a sequence



1. n^{th} term of a sequence

nth term of a sequence is an expression in which

if we put n = 1, we get 1st term,

if we put n = 2, we get 2nd term & so on

"n^{th} term" Very Important

"t_n" $\longrightarrow t_1, t_2, t_3 \ldots\ldots$

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$t_n \rightarrow n^{th}$ terms

$n \in N, Z^+$

$n=1,\ t_1$ ✓

$n=2,\ t_2$ ✓

$n=3,\ t_3$ ✓

$n=4,\ t_4$ ✓

Let's Solve



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Let's Solve

Ex. if $a_n = 2^n$ then find

$a_1 = 2^1 \rightarrow t_1$

$a_2 = 2^2 \rightarrow t_2$

$a_3 = 2^3 \rightarrow t_3$

$a_n = 2^n$

$a_1 a_2 a_3 \cdots a_n$ ✓

$\downarrow$ $\downarrow$

t_n t_n

t_n

a_n

p_n

l_n

$a_1 = 2^1, a_2 = 2^2, a_3 = 2^3$



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Let's Solve

Ex. if $a_n = 2^n$ then find

a_1

a_2

a_3

Solution

$a_n = 2^n$ ✓ → $t_n = 2^n = a_n$

$a_1 = 2^1 = 2$ → $t_1 = a_1 = 2^1$

$a_2 = 2^2 = 4$ → $t_2 = a_2 = 2^2$

$a_3 = 2^3 = 8$ → $t_3 = a_3 = 2^3$

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Let's Solve

$t_n = \longrightarrow$

$t_{20} = (n=20)$

Ex. Find the 20th term of the sequence defined by

$a_n = (n-1)(2-n)(3+n) \rightarrow t_n$

$a_1 = (1-1)(2-1)(3+1) \rightarrow t_1$

$a_2 = (2-1)(2-2)(3+2) \rightarrow t_2$

$a_3 = (3-1)(2-3)(3+3) \rightarrow t_3$

$a_{20} = (20-1)(2-20)(3+20)$

$= 19 \times (-18) \times 23$

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Let's Solve

Ex. Find the 20th term of the sequence defined by

$$a_n = (n-1)(2-n)(3+n)$$

$a_{25} = (25-1)(2-25)(3+25)$

$=$

Solution

given $a_n = (n-1)(2-n)(3+n)$

Putting $n = 20$, we obtain

$$a_{20} = (20-1)(2-20)(3+20)$$

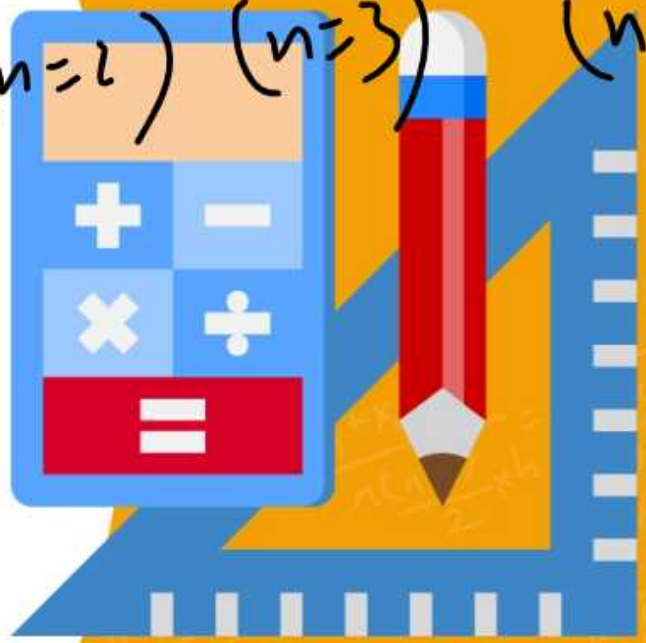
$$a_{20} = -7866$$

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Quiz.

$t_n \rightarrow n^{th}$ term

t_1 $(n=1)$ t_2 $(n=2)$ t_3 $(n=3)$ t_4 $(n=4)$



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Quiz.

Chat Off

Q. If $a_n = \frac{n^2}{2^n}$ then a_7 is

$2^4 \times 2^3 = 16 \times 8$

(A) $\frac{49}{128}$ (B) $\frac{8}{64}$ (C) $\frac{14}{64}$ (D) None

$a_n = \frac{n^2}{2^n}$

$a_7 = \frac{7^2}{2^7} = \frac{49}{128}$

$(n=7)$

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Quiz.

Chat On

Q. If $a_n = \frac{n^2}{2^n}$ then a_7 is

A $\frac{49}{128}$

B $\frac{8}{64}$

C $\frac{14}{64}$

D None

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Quiz.

Q. If $a_n = \frac{n^2}{2^n}$ then a_7 is

$a_n = \frac{n^2}{2^n}$

Solution

$$a_n = \frac{n^2}{2^n}$$

$$a_7 = \frac{7^2}{2^7} = \frac{49}{128}$$

$a_n = \frac{n^2}{2^n}$

$a_7 = \frac{7^2}{2^7} = \frac{49}{128}$ //



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Let's Solve



Let's Solve

$k=3$

$a_{k+2} = a_{k+1} + a_k$

$a_{3+2} = a_{3+1} + a_3$

$a_5 = a_4 + a_3 = 5 + 3 = 8$

Ex. Find 3rd, 4th, 5th terms of sequence defined as:

$$a_1 = 1,\ a_2 = 2 \ \&\ a_{k+2} = a_{k+1} + a_k \ \ \forall\ k \geq 1$$

$k \geq 1$

$a_1 = 1,\ a_2 = 2,$

$a_{k+2} = a_{k+1} + a_k$

$k=1$ $a_{1+2} = a_{1+1} + a_1$

$a_3 = a_2 + a_1$

$a_3 = 2 + 1$

$a_3 = 3$

$a_{k+2} = a_{k+1} + a_k$

$k=2$ $a_{2+2} = a_{2+1} + a_2$

$a_4 = a_3 + a_2$

$a_4 = 3 + 2$

$a_4 = 5$

$a_3 = 3,\ a_4 = 5,\ a_5 = 8$

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Let's Solve

Ex. Find 3rd, 4th, 5th terms of sequence defined as:

$$a_1 = 1,\ a_2 = 2 \ \&\ a_{k+2} = a_{k+1} + a_k \quad \forall\ k \geq 1$$

Solution

Given, $a_1 = 1,\ a_2 = 2$

$$a_{K+2} = a_{K+1} + a_K \quad \forall K \geq 1$$

Let's put $k = 1$

$$a_3 = a_2 + a_1 = 1 + 2 = 3$$

$k = 2$

$$a_4 = a_3 + a_2 = 5$$

$k = 3$

$$a_5 = a_4 + a_3 = 5 + 3 = 8$$

$a_1 = 1,\ a_2 = 2$

$a_{K+2} = a_{K+1} + a_K$

$K = 1$ $\quad a_3 = a_2 + a_1$

$a_3 = 2 + 1 = 3$

$a_{K+2} = a_{K+1} + a_K$

$K = 2$ $\quad a_4 = a_3 + a_2$

$a_4 = 3 + 2 = 5$

$a_{K+2} = a_{K+1} + a_K$

$K = 3$ $\quad a_5 = a_4 + a_3$

$a_5 = 5 + 3$

$= 8$

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Sequence and Series

2. Arithmetic Progression

$t_n = n^2 + n + 1$

$t_1 = 1 + 1 + 1 = 3$

$t_2 = 2^2 + 2 + 1 = 7$

$t_3 = 3^2 + 3 + 1 = 13$

Progression

"Progression"

numbers formed using a "common property"

3, 7, 13, ---

Progression

$t_n = n^3$

1, 8, 27, 64, 125, ----

$t_n = n^2 - n$

Progression, $2^2 - 2$, $3^2 - 3$, $4^2 - 4$, ---

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2.1 Arithmetic Progression

Sequence of the term: $a, a+d, a+2d, \ldots\ldots + a+(n-1)d$

is called an A.P.,

where a → 1st term,

d → Common difference

n^{th} term of an A.P is given by

$$T_n = a + (n-1)\,d$$

$a_1 + (n-1)\,d$

2, 7, 12, 17, 22, ---

Rule → property

Arithmetic progression pattern

$a, a+d, a+2d, a+3d, \ldots$

$a, a-d, a-2d, a-3d$

If sequence has n terms then, last term is also denoted by 'l' sometimes.

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$t_2 - t_1 = t_3 - t_2 = t_4 - t_3 = \cdots$

$= t_n - t_{n-1} =$

$\cdots = 'd'$

Let's Solve



$t_n = a + (n-1)d$

$t_n - t_{n-1} = 'd'$

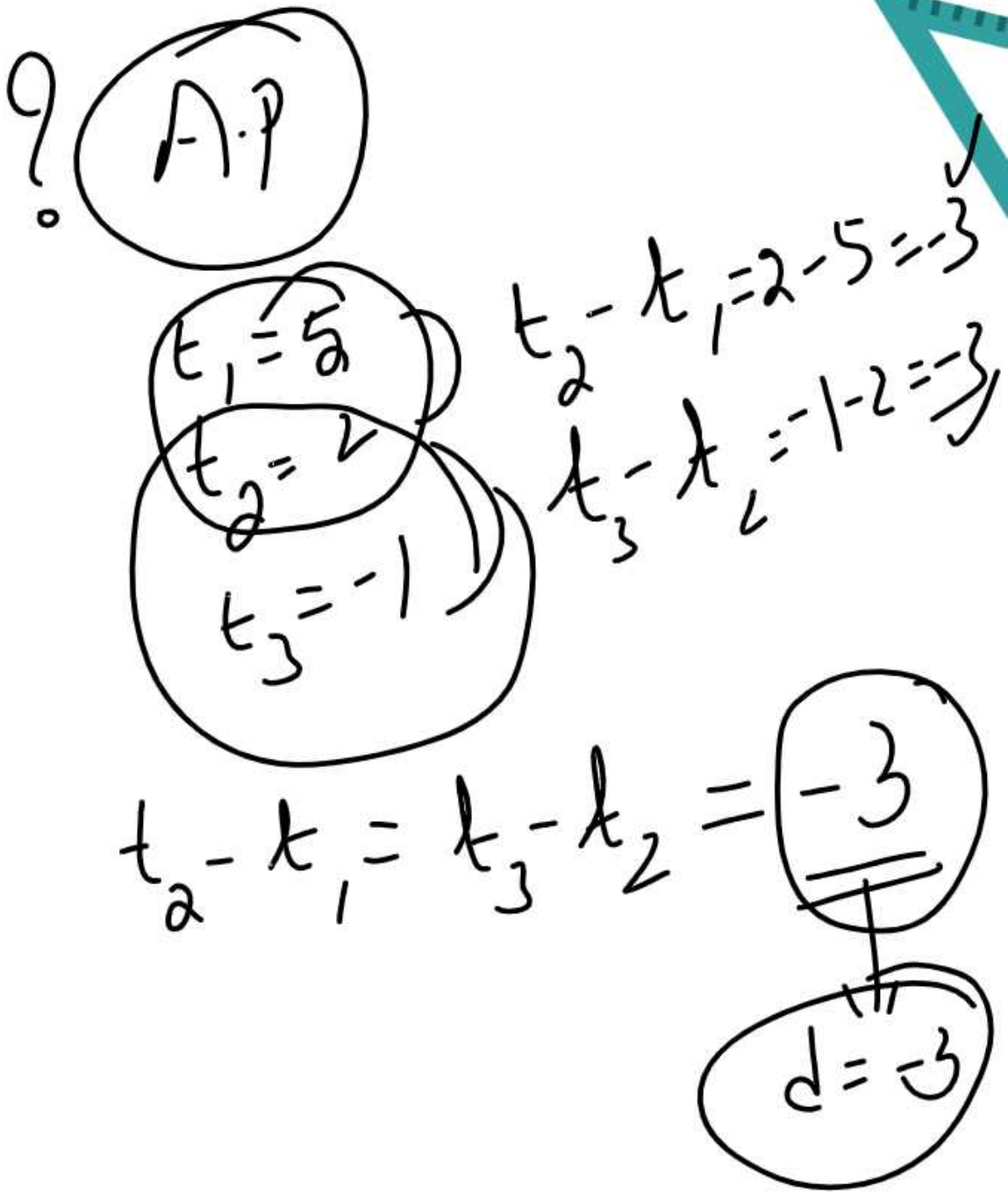
'd' is Common difference

Let's Solve

Ex. Find the nth term of:

(a) $5, 2, -1, ...$

(b) $\log a, \log ab, \log ab^2$



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Let's Solve

Ex. Find the nth term of:

(a) $5, 2, -1, \ldots\ldots$

$t_2 - t_1 = t_3 - t_2 = -3$

$2 - 5 = -1 - 2 = -3$

$d = -3$

$5, 2, -1, - - - -$

$a = 5 \quad d = -3$

$t_n = a + (n-1)d$

$t_n = 5 + (n-1)(-3)$

$t_n = 5 + (-3n + 3)$

$t_n = 8 - 3n$

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Let's Solve

Ex. Find the nth term of:

(a) $5, 2, -1, ...$

Solution

Given, $5, 2, -1, ...$

Since, $2-5=-3$

and $-1-2=-3$

so the sequence is in A. P.

$a = 5$

$d = -3$

$T_n = a + (n-1)d = 5 + (n-1)(-3)$

$= 8 - 3n$

$t_2 - t_1 = t_3 - t_2$

$= t_4 - t_3 = \cdots$

$\cdots = t_n - t_{n-1}$

a = 1st term

Same $\rightarrow$ $d = -3$ Common difference = d

$T_n = a + (n-1)d$

$= 5 + (n-1)(-3)$

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Let's Solve

Ex. Find the nth term of:

(b) $\log a, \log ab, \log ab^2, \ldots$

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Let's Solve

Ex. Find the nth term of:

(b) $\log a, \log ab, \log ab^2, \ldots$

Solution

Given, $\log a, \log ab, \log ab^2, \ldots$

Here, first term (let A) $= \log a$

Common difference d $= \log ab - \log a$

$= \log b$

$$\therefore T_n = \log a + (n-1)(\log b)$$

$$= \log\left(\frac{a}{b}\right) + n(\log b)$$

$$= \log\left(ab^{n-1}\right)$$

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Let's Solve

$t_n = a + (n-1)d$

Ex. If 6th & 11th term of an AP is 17 & 32 then find 15th term

$t_6 = 17, \quad t_{11} = 32 \quad t_{15} = ?$

$t_6 = a + (6-1)d = 17$

$t_6 = a + 5d = 17 \quad -(1)$

$t_{11} = a + (11-1)d = 32$

$t_{11} = a + 10d = 32 \quad -(2)$

$$\begin{cases} a + 5d = 17 \\ a + 10d = 32 \quad (-) \end{cases}$$

$-5d = -15$

$d = 3$

$a + 5d = 17$

$a + 5(3) = 17$

$a = 17 - 15 = 2$

$a = 2$

$t_{15} = a + (15-1)d$

$= a + 14d$

$= 2 + 14(3)$

$= 44$

Observation.

Try to focus on 'a' & 'd' only. AP is all about 'a' & 'd'

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Let's Solve

$t_{15} = 3(15) - 1 = 45 - 1 = 44$

Ex. If 6th & 11th term of an AP is 17 & 32 then find 15th term

Solution

Given, $T_6 = 17 = a + 5d$ ——(1)

$T_{11} = 32 = a + 10d$ .....(2)

$a + 5d = 17$

$a + 10d = 32$ (-1)

$-5d = -15$

$d = 3$

Solving (1) and (2), we get

$5d = 15$

$\Rightarrow d = 3$

$a = 2, d = 3$

Thus $a = 17 - 15 = 2$

$\therefore T_{15} = a + 14d = 2 + 14(3)$

$= 2 + 42 = 44$

$t_n = 2 + (n-1)3$

$t_n = 2 + 3n - 3$

$t_n = 3n - 1$

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Quiz.



Quiz.

Chat Off

$$t_n = a + (n-1)d$$

Q. If the 9th term of an A.P. is 35 and 19th term is 75, then its 20th terms will be

$t_n = 4n - 1 \longrightarrow t_{20} = 4(20) - 1 = 79$

A 78 B 79 C 80 D 81

$t_9 = a + (9-1)d = 35 \longrightarrow a + 8d = 35$ — (1)

$t_{19} = a + (19-1)d = 75 \longrightarrow a + 18d = 75$ — (2)

$(-)$

$-10d = -40$

$d = 4$

(1) $\rightarrow a + 8d = 35$

$d = 4$

$a + 8(4) = 35$

$a = 3$

$a = 3, d = 4$

$t_n = a + (n-1)d$

$t_n = 3 + (n-1)4$

$t_n = 4n - 1$

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Quiz.

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Q. If the 9th term of an A.P. is 35 and 19th term is 75, then its 20th terms will be

A 78

B 79 ✓

C 80

D 81

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Quiz.

Q. If the 9th term of an A.P. is 35 and 19th term is 75, then its 20th terms will be

Solution

$T_9 = a + 8d = 35$ and $T_{19} = a + 18d = 75$

Solving the equations, we get $d = 4$ and $a = 3$.

Hence 20th term of A.P.

$$a + 19d = 3 + 19 \times 4 = 79$$

$t_n = a + (n-1)d$

$t_n = 3 + (n-1)4$

$t_n = 4n - 1$

$t_{20} = 4(20) - 1 = 79$

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Quiz.

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$$t_n = a_n = A + (n-1)D$$

A: 1st term, D: Common diff

Q. In an A.P. if $a_2 + a_5 - a_3 = 10$ & $a_2 + a_9 = 17$ then find a_{20}

A 6 **B** −6 **C** 8 **D** −10

$a_2 + a_5 - a_3 = 10$; $a_2 + a_9 = 17$

$a_2 = A + (2-1)D = A + D$

$a_5 = A + (5-1)D = A + 4D$

$2A + 5D$

$-\{a_3 = A + (3-1)D = A + 2D\}$

$A + 3D = 10$ — (1)

$(A + D) + (A + 8D) = 17$

$2A + 9D = 17$ — (2)

$A + 3D = 10 \times 2$

$2A + 9D = 17$

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$A + 3D = 10$ $\times 2$ ✓

$2A + 9D = 17$

$2A + 6D = 20$

$2A + 9D = 17$

(-1) (-1)

$-3D = 3$

$D = -1$

$2A + 9D = 17$

$D = -1$ ✓

$2A + 9(-1) = 17$

$2A = 17 + 9$

$A = 13$

$t_n = 13 + (n-1)(-1)$

$t_n = 13 - n + 1$

$t_n = 14 - n$

$t_{20} = 14 - 20 = -6$

Quiz.

Chat On

Q. In an A.P. if $a_2 + a_5 - a_3 = 10$ & $a_2 + a_9 = 17$ then find a_{20}

A 6

B - 6

C 8

D - 10



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Quiz.

Q. In an A.P. if $a_2 + a_5 - a_3 = 10$ & $a_2 + a_9 = 17$ then find a_{20}

Solution

Given, $a_2 + a_5 - a_3 = 10$

$\therefore (a+d) + (a+4d) - (a+2d) = 10$

$a + 3d = 10 \quad \ldots.(1)$

Also, $a_2 + a_9 = 17$

$(a+d) + (a+8d) = 17$

$2a + 9d = 17 \quad \ldots.(2)$

Solving (1) and (2)

$3d = -3$

$d = -1$

Thus $a = 10 + 3 = 13$

Thus $a_{20} = a + 19d = 13 - 19 = -6$

$a_n = 13 + (n-1)(-1)$

$a_n = 14 - n$

$a_{20} = 14 - 20 = -6$

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Let's Solve



Let's Solve

$t_n = a + (n-1)d$

Ex. If in an A.P. $a_p = \frac{1}{q}$ & $a_q = \frac{1}{p}$ then find a_{pq}

$a_p = \frac{1}{q}$

$t_p \; a_p = a + (p-1)d = \frac{1}{q} \quad -(1)$

$t_p \; a_q = a + (q-1)d = \frac{1}{p} \quad -(2)$

$a = ?, \; d = ?$ (-)

$(p-1)d - (q-1)d = \frac{1}{q} - \frac{1}{p}$

$d(p-1-q+1) = \frac{p-q}{qp}$

$d(p-q) = \frac{p-q}{qp}$

$d = \frac{1}{qp}$

$a + (p-1)d = \frac{1}{q}$

$a + (p-1)\frac{1}{qp} = \frac{1}{q}$

$a + \frac{1}{q} - \frac{1}{qp} = \frac{1}{q}$

$a = +\frac{1}{pq}$

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$$a_{(pq)} = a + (pq-1)d$$

$$a = \frac{1}{pq} \qquad d = \frac{1}{pq}$$

$$a_{pq} = \frac{1}{pq} + (pq-1)\frac{1}{pq}$$

$$= \frac{1}{pq} + 1 - \frac{1}{pq}$$

$$a_{pq} = 1$$

Let's Solve

Ex. If in an A.P. $a_p = \frac{1}{q}$ & $a_q = \frac{1}{p}$ then find a_{pq}

Solution

Given,

$$a_p = \frac{1}{q} = a + (p-1)d \quad ...(1)$$

$$a_q = \frac{1}{p} = a + (q-1)d \quad ...(2)$$

Solving (1) and (2)

$$d = \frac{1}{pq}$$

$$a = \frac{1}{pq}$$

$$\therefore a_{pq} = a + (pq-1)d$$

$$= \frac{1}{pq} + (pq-1)\frac{1}{pq}$$

$$= \frac{1}{pq} + 1 - \frac{1}{pq}$$

$$a_{pq} = 1$$

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Let's Solve

$t_n = a + (n-1)d$

Ex. If $(x+1), 3x$ and $(4x+2)$ are first three terms of an AP then its 5th term is

$t_5 = 4 + (5-1)5 = 24$

$t_n = a + (n-1)d = 4 + (n-1)5$

A 14 **B** 19 **C** 24 **D** 28

$t_1 = x+1,\ t_2 = 3x,\ t_3 = 4x+2$

$a = x+1;\ a+d = 3x;\ a+2d = 4x+2$

$a = 3+1 = 4$

$(x+1) + d = 3x$

$d + 1 = 2x$

$d = 2x - 1$

$(x+1) + 2(2x-1) = 4x+2$

$x + 1 + 4x - 2 = 4x + 2$

$x - 1 = 2$

$x = 3$

$d = t_2 - t_1$
$= 3x - (x+1)$
$= 2x - 1$
$= 2(3) - 1$
$= 5$

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Let's Solve

Ex. If $(x+1), 3x$ and $(4x+2)$ are first three terms of an AP then its 5th term is

A 14

B 19

C 24

D 28

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Let's Solve

Ex. If $(x+1), 3x$ and $(4x+2)$ are first three terms of an AP then its 5th term is

Solution

t_1 t_2 t_3

$\because\ x+1, 3x$ and $4x+2$ are in A.P.

$\therefore 2 \times 3x = x+1+4x+2$

$\Rightarrow x = 3$

$\therefore a = x+1 = 4$

$d = 3x-(x+1) = 5$

$\therefore t_5 = a+4d = 4+4\times 5 = 24$

$t_1 = a = x+1$

$t_2 = a+d = 3x$

$t_3 = a+2d = 4x+2$

$x = 3$

$a = x+1 = 4$

$t_1 = 4$, $t_2 = 9$, $t_3 = 14$ $\Big\}$ $d = 5$

$a = 4,\ d = 5$

$t_n = a+(n-1)d = 4+(n-1)5$

$t_5 = 4+(5-1)5 = 24$

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Quiz.



Quiz.

Chat Off

"Reverse"

Q. The 19th term from the ~~end of~~ the sequence $2+6+10+\ldots+86$ is

A 6 **B** 10 **C** 14 **D** 18

86, - - - - 10, 6, 2

-4 -4

$a = 86, \quad d = -4$

$t_n = 86 + (n-1)(-4)$

$t_{19} = 86 + (19-1)(-4)$

$= 86 - 72$

$t_{19} = 14$

$d = 4$

-1

Reverse

$d = -4$



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Quiz.

Chat On

Q. The 19th term from the end of the sequence $2+6+10+...+86$ is

A 6

B 10

C 14

D 18



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Quiz.

Q. The 19th term from the end of the sequence $2+6+10+...+86$ is

Solution

On reversing the order of terms we get

$86, 82, 78, ..., 10, 6, 2$

19th term from the end for given series will be same as 19th term from the beginning in above Progression

Hence required term is

$$a_{19} = 86 + (19-1)(-4)$$
$$= 14$$

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Let's Solve



Let's Solve

(*) $a_n = a_1 + (n-1)d$

$a_n - a_1 = (n-1)d$

Ex. For an A.P. (a_n), where $a_i > 0 \ \forall i$ show that:

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \frac{1}{\sqrt{a_3}+\sqrt{a_4}} = \frac{3}{\sqrt{a_1}+\sqrt{a_4}}$$

$$LAS = \frac{(\sqrt{a_2}-\sqrt{a_1})}{(\sqrt{a_2}-\sqrt{a_1})(\sqrt{a_2}+\sqrt{a_1})} + \frac{(\sqrt{a_3}-\sqrt{a_2})}{(\sqrt{a_3}+\sqrt{a_2})(\sqrt{a_3}-\sqrt{a_2})} + \frac{(\sqrt{a_4}-\sqrt{a_3})}{(\sqrt{a_4}+\sqrt{a_3})(\sqrt{a_4}-\sqrt{a_3})}$$

$$= \frac{\sqrt{a_2}-\sqrt{a_1}}{a_2-a_1} + \frac{\sqrt{a_3}-\sqrt{a_2}}{a_3-a_2} + \frac{\sqrt{a_4}-\sqrt{a_3}}{a_4-a_3}$$

$$= \frac{\sqrt{a_2}-\sqrt{a_1}+\sqrt{a_3}-\sqrt{a_2}+\sqrt{a_4}-\sqrt{a_3}}{d}$$

Hint

Rationalize

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$$\frac{1}{\sqrt{a_2}+\sqrt{a_1}} + \frac{1}{\sqrt{a_3}+\sqrt{a_2}} + \frac{1}{\sqrt{a_4}+\sqrt{a_3}} =$$

$$\Rightarrow \frac{\sqrt{a_2}-\sqrt{a_1}}{(\sqrt{a_2}-\sqrt{a_1})(\sqrt{a_2}+\sqrt{a_1})} + \frac{\sqrt{a_3}-\sqrt{a_2}}{(\sqrt{a_3}-\sqrt{a_2})(\sqrt{a_3}+\sqrt{a_2})} + \frac{\sqrt{a_4}-\sqrt{a_3}}{(\sqrt{a_4}-\sqrt{a_3})(\sqrt{a_4}+\sqrt{a_3})}$$

$$\Rightarrow \frac{\sqrt{a_2}-\sqrt{a_1}}{a_2-a_1} + \frac{\sqrt{a_3}-\sqrt{a_2}}{a_3-a_2} + \frac{\sqrt{a_4}-\sqrt{a_3}}{a_4-a_3}$$

$$\Rightarrow \frac{\cancel{\sqrt{a_2}}-\sqrt{a_1}+\cancel{\sqrt{a_3}}-\cancel{\sqrt{a_2}}+\sqrt{a_4}-\cancel{\sqrt{a_3}}}{d} = \frac{\sqrt{a_4}-\sqrt{a_1}}{d}$$

$$\frac{\sqrt{a_4}-\sqrt{a_1}}{d} = \frac{\left(\sqrt{a_4}-\sqrt{a_1}\right)\times\left(\sqrt{a_4}+\sqrt{a_1}\right)}{d\times\left(\sqrt{a_4}+\sqrt{a_1}\right)}$$

$$= \frac{a_4 - a_1}{d\left(\sqrt{a_4}+\sqrt{a_1}\right)}$$

$$a_n = a_1 + (n-1)d$$

$n = 4$

$a_4 = a_1 + 3d$

$a_4 - a_1 = 3d$

$$= \frac{3\cancel{d}}{\cancel{d}\left(\sqrt{a_4}+\sqrt{a_1}\right)} = \frac{3}{\sqrt{a_4}+\sqrt{a_1}}$$

Hence proved

Let's Solve

Ex. For an A.P. (a_n), where $a_i > 0 \ \forall i$ show that:

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+\frac{1}{\sqrt{a_3}+\sqrt{a_4}}=\frac{3}{\sqrt{a_1}+\sqrt{a_4}}$$

Solution

Let terms in A.P. be $a_1, a_2, a_3, \ldots\ldots a_n$

$$\therefore a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \ldots. = a_n - a_{n-1} = d$$

Now,

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+\frac{1}{\sqrt{a_3}+\sqrt{a_4}}$$

Rationalize

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Let's Solve

Ex. For an A.P. (a_n), where $a_i > 0 \ \forall i$ show that:

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+\frac{1}{\sqrt{a_3}+\sqrt{a_4}}=\frac{3}{\sqrt{a_1}+\sqrt{a_4}}$$

Solution

$$\frac{\sqrt{a_1}-\sqrt{a_2}}{a_1-a_2}+\frac{\sqrt{a_2}-\sqrt{a_3}}{a_2-a_3}+\frac{\sqrt{a_3}-\sqrt{a_4}}{a_3-a_4}$$

$$\Rightarrow \quad \frac{1}{-d}[\sqrt{a_1}-\sqrt{a_2}+\sqrt{a_2}-\sqrt{a_3}+\sqrt{a_3}-\sqrt{a_4}]$$

$$\Rightarrow \quad -\frac{1}{d}[\sqrt{a_1}-\sqrt{a_4}]$$

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Let's Solve

Ex. For an A.P. (a_n), where $a_i > 0 \ \forall i$ show that:

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+\frac{1}{\sqrt{a_3}+\sqrt{a_4}}=\frac{3}{\sqrt{a_1}+\sqrt{a_4}}$$

Solution

$$\Rightarrow -\frac{1}{d}\left[\frac{a_1-a_4}{\sqrt{a_1}+\sqrt{a_4}}\right]$$

$$\Rightarrow -\frac{1}{d}\left[\frac{a_1-(a_1+3d)}{\sqrt{a_1}+\sqrt{a_4}}\right]$$

$$\Rightarrow -\frac{1}{d}\left[\frac{-3d}{\sqrt{a_1}+\sqrt{a_4}}\right]=\frac{3}{\sqrt{a_1}+\sqrt{a_4}}$$

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Let's Solve

Ex. Find 10th common term in: $3,7,11,15,19,....$ & $1,6,11,16,21,....$

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Let's Solve

Ex. Find 10th common term in: $3,7,11,15,19,....\&1,6,11,16,21,....$

Solution

Given two series in A.P.

$3,7,11,15,19,23,27,31,35,...$

and $1,6,11,16,21,26,31,36,....$

So common terms will also form an A.P.

$11,31,51,....$ with common difference d = 20

Thus $T_{10} = 11+(10-1)20$

$=11+180$

$=191$

Also, common difference of common terms which forms an A.P. is always LCM of common difference of respective series.

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Quiz.



Quiz.

Chat Off

Q. If a, b, c, d, e, f are in A.P., then the value of $e-c$ will be

A $2(c-a)$ B $2(f-d)$ C $2(d-c)$ D $d-c$

$t_1 = a = a$

$t_2 = b = a+d$

$t_3 = c = a+2d$

$t_4 = d = a+3d$

$t_5 = e = a+4d$

$t_6 = f = a+5d$

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Quiz.

Chat On

Q. If a, b, c, d, e, f are in A.P., then the value of $e - c$ will be

A $2(c - a)$ B $2(f - d)$ C $2(d - c)$ D $d - c$



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Quiz.

Q. If a,b,c,d,e,f are in A.P., then the value of $e-c$ will be

Solution

a,b,c,d,e,f are in A.P.

So, $b-a=c-b=d-c=e-d=f-e=K$

where K is a common difference

Now, $d-c=e-d \Rightarrow e+c=2d$

$e-c+2c=2d \Rightarrow e-c=2(d-c)$



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Quiz.

Chat Off

JEE Mains 2020

Q. The common difference of the A.P. $b_1, b_2, ..., b_m$ is 2 more than the common difference of A.P. $a_1, a_2, ..., a_n$. If $a_{40} = -159, a_{100} = -399$ And $b_{100} = a_{70}$, then b_1 is equal to :

A 81

B -127

C -81

D 127

Vedantu

Quiz.

Chat On

JEE Mains 2020

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Vedantu

Quiz.

JEE Mains 2020

Q. The common difference of the A.P. $b_1, b_2, ..., b_m$ is 2 more than the common difference of A.P. $a_1, a_2, ..., a_n$. If $a_{40} = -159, a_{100} = -399$ And $b_{100} = a_{70}$, then b_1 is equal to :

Solution

Let common difference of series

$a_1, a_2, a_3, ..., a_n$ be d.

$\because\ a_{40} = a_1 - 39d = -159 \qquad ...(i)$

and $a_{100} = a_1 + 99d = -399 \qquad ...(ii)$

From equations (i) and (ii),

$d = -4$ and $a_1 = -3$

Since, the common difference of $b_1, b_2, ..., b_n$ is 2 more

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Quiz.

JEE Mains 2020

Q. The common difference of the A.P. $b_1, b_2, ..., b_m$ is 2 more than the common difference of A.P. $a_1, a_2, ..., a_n$. If $a_{40} = -159, a_{100} = -399$ And $b_{100} = a_{70}$, then b_1 is equal to :

Solution

then common difference of $a_1, a_2, ..., a_n$

$\therefore$ common difference of $b_1, b_2, b_3, ...$ is (- 2)

$\because b_{100} = a_{70}$

$\Rightarrow b_1 + 99(-2) = (-3) + 69(-4)$

$\Rightarrow b_1 = 198 - 279 \Rightarrow b_1 = -81$

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Summary.



Summary.

1. n^{th} term of a Sequence
2. Arithmetic Progression
3. n^{th} term of Arithmetic Progression

$$T_n = a + (n-1)d$$

$$t_n - t_{n-1} = d$$



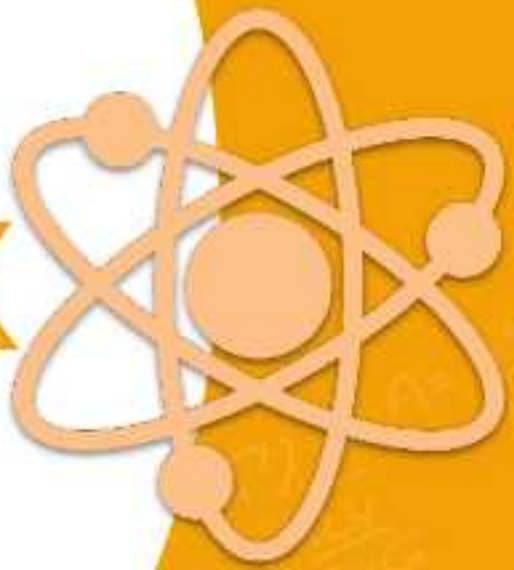
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Tatva Ex-1 HW Q. No's

Q. No's.								
1	2	3	4	5	60	61	62	63

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Feedback



Thanks.

